

Does topology causally influence physics?

Winding numbers as difference-makers in a two-dimensional Bose gas: thought experiments, a machine-checked chain, and an energy-matched intervention

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SocrateAI-Scientific-QuantumFluids

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Abstract

We ask, in the interventionist sense of Woodward and Pearl, whether the topological charge of a superfluid phase field – the winding number of the phase around a loop – is a *cause* of what the fluid does, or only a label of its microstate. The question splits in two. First, is the integer stable enough to carry an intervention: is it conserved, and at what cost is it changed? Second, at a fixed thermodynamic state, is it a difference-maker for the observables? We answer the first part with machine-checked mathematics (Lean 4, Mathlib, two independent kernels): the discrete winding of a sampled phase loop is conserved by every continuous evolution that avoids a phase slip, a change of winding forces a phase slip, on an XY ring the slip costs a positive energy barrier for rings of at least ten sites (and the bound fails at nine), the winding seen at scale R is the net charge inside R so that a bound pair is invisible from any enclosing scale, and – closing a gap left open in this project’s earlier work – the sampled principal-branch loop sum equals the topological degree of the continuum phase for every fine enough sampling. We answer the second part with a pre-registered, energy-matched intervention in a projected Gross–Pitaevskii classical field: from equilibrated states we inject four free vortex–antivortex pairs at fixed particle number, momentum and energy, against a control arm that receives the same energy as phonons. On all three base states the condensate fraction falls by 0.49–0.73 in the vortex arm and by at most 0.035 in the phonon arm, and the coherence exponent η grows by a factor 9–44 against at most 1.2: the same energy, delivered as topology, has a 15–60 $\times$ larger effect. The pre-registered composite claim is nevertheless not made, because its thermometer clause fails: the vortex arm reads *colder*, by 5–16%, than the phonon arm and than the untouched base, which we read, post hoc, as the topological configuration drawing heat from the bath – a difference with the wrong sign to explain the effect. At the lowest temperature the eight imprinted vortices survive the whole run without one annihilation. Heating the equilibrated states step by step locates the stiffness jump at $T_{\text{BKT}} = 0.821$ with $\eta = 0.31$ there, confirms the spin-wave duality $\eta n_s \lambda_T^2 = 1$ to 12–27% on eleven energies in two box sizes, and shows that a quench state with the same number of vortices as an equilibrated one can have half its stiffness. None of the physics is new; what is new is the chain, each link either proved or measured under a pre-registered criterion, and the explicit list of what the chain does not establish.

1 The question, made precise

“Topology influences physics” can mean three things, and they must be kept apart.

1. **Descriptor.** The winding number w summarises a microstate. Every function of the microstate does; nothing follows.
2. **Constraint.** Configuration space is partitioned into sectors labelled by w , and the dynamics cannot move continuously between them. This is a statement about the space, independent of the Hamiltonian.

3. **Difference-maker.** An intervention that changes w while holding the thermodynamic state fixed changes the observables. This is causation in the interventionist sense [1, 2].

The obstacle to the third sense is that w supervenes on the microstate: one cannot change w and nothing else. What one can do is compare *matched pairs*: two microstates in the same macrostate (E, N, P) , one with the topological label and one without, and ask whether the observables differ. That is the experiment of Section 5. Whether the comparison is fair is then the whole question, and it is answered by making the two arms equally arbitrary: both are the same equilibrated base state plus the same energy, delivered once as vortex flow and once as phonons.

The second sense is what makes the third possible. If w could jump freely, no intervention on it would leave a trace, and the matched pair would decorrelate at once. So the order of the argument is: prove the constraint (Section 4), then measure the difference (Section 5).

2 Thought experiments

A. The rotating ring. A superfluid ring is given one turn of phase and left alone. The current persists for hours in helium and for tens of seconds in a toroidal condensate [3]. No thermodynamic variable remembers the intervention; the integer does. Topology acts as memory. This is the constraint sense, and it is what Theorems 2–3 formalise.

B. The small-pair trap. A vortex and an antivortex a distance a apart. A loop around one sees $w = \pm 1$; any loop of size $R \gg a$ enclosing both sees 0. From far away the pair is a phonon. The causal variable is therefore not “the number of vortices” but the winding seen at scale R , $W(R)$, and the Kosterlitz–Thouless transition [4] is where $W(R)$ stops vanishing at large R . Theorem 4 makes this exact on a lattice. It also explains a failure in this project’s previous round: a pairing statistic that counts pairs at every scale, grid noise included, saturates and cannot see unbinding.

C. Aharonov–Bohm. The electron never enters the flux and its fringes move [5]. Here the topology of space is not the cause; it is the *channel* through which a cause acts at a distance. The two roles must not be confused.

D. Laughlin’s cylinder. Threading one flux quantum through a Hall cylinder returns the Hamiltonian to itself, so an integer number of electrons has been pumped and $\sigma_{xy} \in \mathbb{Z} e^2/h$ [6]. Topology fixes an integer that disorder cannot move – the constraint sense at its most spectacular.

E. The two boxes. Two boxes with the same E, N, P . One receives four free vortex–antivortex pairs, the other the same energy as phonons. If the vortex box has a lower condensate fraction and a faster loss of phase coherence while a thermometer reads the two boxes alike, then w is a difference-maker at fixed thermodynamic state. This is the experiment we run.

3 The physical argument in four links

1. **Integrality.** ψ is single-valued, so $\oint \nabla \theta \cdot d\ell = 2\pi w$.
2. **Continuity implies conservation.** An integer-valued continuous function of time is constant. The only exit is $\psi = 0$ somewhere on the loop: a vortex core crosses it, Anderson’s phase slip [7].
3. **Barrier.** Passing through $\psi = 0$ costs energy, so slips are activated and the protection is exponential in the barrier over temperature [8].

4. **Coupling.** The integer enters the observables: circulation $\Gamma = w\kappa$, the persistent current, and – for free vortices in two dimensions – the exponential decay of g_1 that replaces the algebraic one [4, 9].

Noether charges come from a symmetry of the Hamiltonian; topological charges come from the connectivity of configuration space, $\pi_1(U(1)) = \mathbb{Z}$, and are conserved by continuity alone. Topology does not push. It partitions the state space into sectors the dynamics cannot connect continuously, and a conserved sector label is exactly the kind of variable an intervention can act on.

4 What is proved

All statements below are theorems in Lean 4 against the pinned Mathlib of this project [10], with axiom footprint {propxt, Classical.choice, Quot.sound}, accepted by the Lean kernel and by the independent `nanoda` kernel through Comparator, and each accompanied by negative controls – weakened statements whose proofs must, and do, fail. We state them in words; the files carry the exact forms.

4.1 Conservation and the slip (`TopologicalProtection.lean`, 13 theorems)

Let $\theta_0, \dots, \theta_n = \theta_0$ be phases read at the sample points of a closed loop, and $\text{pd}(a, b) \in (-\pi, \pi]$ the principal difference. The loop sum $S = \sum_i \text{pd}(\theta_i, \theta_{i+1})$ is 2π times an integer (earlier work).

Theorem 1 (stability). *A perturbation δ_i of the phases, closed ($\delta_n = \delta_0$), leaves S unchanged provided no edge is pushed through the branch point: $\text{pd}(\theta_i, \theta_{i+1}) + \delta_{i+1} - \delta_i \in (-\pi, \pi]$ for all i . In particular any perturbation with $|\delta_i| < \varepsilon$ leaves the winding of a loop whose edge steps are all below $\pi - 2\varepsilon$ unchanged.*

Theorem 2 (continuous-time protection). *If each $\theta_i(t)$ is continuous, the loop stays closed, and no edge step reaches π on $[t_0, t_1]$, then $S(t_1) = S(t_0)$. Contrapositive: a change of winding forces some edge step to reach π at some intermediate time – a phase slip.*

The proof is the intermediate value theorem applied to an integer-valued continuous function; the continuity of pd away from the branch point is Mathlib’s `continuousAt_toIocMod`. The circulation $\Gamma = (\hbar/m)S$ is conserved under the same hypothesis: a discrete Kelvin theorem whose only ingredient is topology.

Theorem 3 (mountain pass and barrier). *On the XY ring $E = -J \sum_i \cos(\theta_{i+1} - \theta_i)$, any continuous evolution that changes the winding passes through a state with $E \geq -(n-2)J$. From the uniformly twisted state of winding one, $E = -nJ \cos(2\pi/n)$, the barrier is therefore at least $nJ \cos(2\pi/n) - (n-2)J \geq 2J - 2\pi^2 J/n > 0$ for every $n \geq 10$.*

The threshold is sharp as stated: at $n = 9$, $9 \cos(2\pi/9) = 6.894 < 7$ and the inequality is false; that false statement is one of the negative controls.

4.2 Scale (`ScaleResolvedWinding.lean`, 6 theorems)

A region is a finite family of plaquettes with four corner phases each; interior edges come in reversed pairs, boundary edges are the rest.

Theorem 4 (discrete Stokes for windings). *If no interior edge sits at the branch point, the oriented sum over the boundary edges equals the sum of the plaquette loop sums, hence 2π times the net charge inside.*

Corollary 5. *A region whose charges sum to zero – a bound pair, at any separation – has boundary winding 0 whatever its size. A region containing exactly one charged plaquette has boundary winding equal to that charge, whatever its size.*

The winding at scale R is the net charge inside R . This is thought experiment B.

4.3 Discrete equals continuum (ContinuumWinding.lean, 9 theorems)

Theorem 6 (the discrete winding is the degree). *Let $\gamma : [0, 1] \rightarrow \mathbb{R}/2\pi\mathbb{Z}$ be a continuous closed loop of angles. There is an integer w and a threshold n_0 such that for every $n \geq n_0$ and every choice of real representatives θ_i of the sampled angles $\gamma(i/n)$, $\sum_{i < n} \text{pd}(\theta_i, \theta_{i+1}) = 2\pi w$.*

Corollary 7 (the vortex detector is correct). *For a continuous, nowhere-vanishing, closed loop of field values $c : [0, 1] \rightarrow \mathbb{C}$, the principal-branch loop sum of $\arg c(i/n)$ equals $2\pi w$ for all $n \geq n_0$.*

The lift is Mathlib’s path lifting through the covering $\mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$, the threshold comes from Heine–Cantor, and the continuity of $\arg$ as an angle away from zero is Mathlib’s `continuousAt_arg_coe_angle`. The hypothesis $c(t) \neq 0$ is where the phase slip lives in the continuum: at a zero of ψ on the loop the angle loop, and with it the lift and the degree, cease to exist.

4.4 What is not proved

The argument principle – that the degree along a loop counts the zeros inside it – is not in the pinned Mathlib for general continuous maps; Theorem 4 is the discrete substitute. The threshold n_0 is existential; a quantitative one needs a modulus of continuity of ψ , i.e. a bound on $|\nabla\theta|$ away from cores. That the twisted state minimises the energy in its sector is not proved. And link 4, the coupling of the integer to condensate and coherence, is not a theorem at all; it is what Section 5 measures.

5 The intervention

5.1 Design (pre-registered)

The model is the projected Gross–Pitaevskii equation on a $64\xi \times 64\xi$ torus ($\hbar = m = g = n = 1$, 128^2 grid, cutoff $k_{\max}/2$), integrated by an integrating-factor RK4 whose seven known answers (norm, energy, momentum, plane wave, Bogoliubov dispersion, an independent integrator, two-window equipartition) were passed before any thermal run [11, 12]. The bases are three states equilibrated for $t = 4000$: two at $e = 0.90$ ($T \approx 0.45$, condensate ≈ 0.63) and one at $e = 0.60$ ($T \approx 0.11$, condensate 0.91).

From each base, arm V imprints four vortex–antivortex pairs of separation $L/4$ with the exact doubly periodic phase (a product of Jacobi ϑ_1 factors, single-valued on the torus when $\sum q_j = 0$ and $\sum q_j \mathbf{r}_j = 0$), a core factor, projection, and renormalisation to the base norm. Arm P adds a random phonon field on $|k| \in [0.2, 1] k_{\text{cut}}$, renormalised, with amplitude bisected until $E_P = E_V$. Arm 0 continues the base untouched. Controls before any arm was run: the detector finds exactly the eight imprinted vortices with the right charges; the boundary phase jump of the imprint is 0 and that of a deliberately non-neutral configuration is $2\pi \cdot 3/L = 0.294524$, as predicted; $|E_V - E_P|/E \leq 2 \times 10^{-16}$; $|P|/N \leq 0.025 \cdot 2\pi/L$.

Pre-registered predictions, compared over $t \in [100, 300]$: (I1) the condensate fraction is lower in V by at least 0.15; (I2) the two-window thermometer agrees within 5%; (I3) the W_1 dipole-matching statistic Q is higher in V by at least 0.2; (I4) the algebraic exponent η of g_1 is at least $1.5\times$ larger in V. A pre-data amendment fixed the reading of I2: arm P heats the bath the thermometer reads, so $T_V < T_P$ by up to the surplus fraction (1.5% at $e = 0.90$, 3.3% at $e = 0.60$) is bookkeeping, not refutation. The causal effect size to be reported is the effect per unit energy: the same ΔE in the two arms, compared on condensate and η .

5.2 Results of the intervention

base	I1: $c_P - c_V \geq 0.15$	I2: $ T_V - T_P /T_P \leq 0.05$	I3: $Q_V - Q_P \geq 0.2$	I4: $\eta_V/\eta_P \geq 1.5$
$e = 0.60$	0.734 pass	0.161 fail ($T_V < T_P$)	not evaluable [†]	43.7 pass
$e = 0.90$, s11	0.565 pass	0.055 fail ($T_V < T_P$)	0.656 pass	10.7 pass
$e = 0.90$, s12	0.488 pass	0.092 fail ($T_V < T_P$)	0.553 pass	8.6 pass

Table 1: Pre-registered verdicts over $t \in [100, 300)$. [†]Arm P carries no vortices at $e = 0.60$, so Q_P is undefined. Window means, V/P: condensate 0.168/0.902, 0.073/0.638, 0.139/0.627; η : 0.98/0.022, 1.08/0.10, 0.94/0.11 (V exponential, P algebraic on every base). Arm 0: $T = 0.462, 0.463$; condensate 0.647, 0.662.

The pre-registered composite claim – “at fixed E, N, P and measured T , the topology sets condensate and coherence”, requiring I1, I2 and I4 on all three bases – is **not made**: I2 fails on all three. It fails, moreover, in none of the three ways the pre-data amendment had anticipated: $T_V < T_P$ by 5.5%, 9.2% and 16%, more than the energy surplus of 1.5–3.3%.

Post hoc. Two explanations were tested and one survives. The coherent spectrum of the vortex cores does not reach the thermometer window (its occupation there is $500\text{--}2000\times$ below the thermal one): refuted. Doppler smearing by the vortex flow: the two thermometer windows disagree alike in V and P: inconclusive. Against the untouched arm, P is warmer by +1.5 to +3.5%, as the amendment predicted, and V is *colder* by -4% . With the total energy conserved to 2×10^{-7} , the bath’s lost heat went into the vortex system: V carries 14–16 vortices against 8 imprinted plus ≈ 5 thermal, and its condensate is scrambled into low- k quasi-condensate structure. The topological configuration cools the bath.

That cooling has the wrong sign to explain the effect. A colder gas carries more condensate; V carries 5–10 $\times$ less. The difference between the arms is therefore not a temperature effect, and the conclusion that survives is stronger than the one pre-registered, though it is post hoc: at fixed E, N and P , and against the temperature gradient, the topological label sets the condensate and the coherence.

Effect per unit energy. The same ΔE in both arms. Relative to arm 0 at $e = 0.90$, the condensate changes by -0.574 and -0.523 in V and by -0.009 and -0.035 in P; η by $\times 10.9$ and $\times 10.4$ in V and $\times 0.99$ and $\times 1.2$ in P. Delivered as topology, the same energy has a 15–60 $\times$ larger effect on the condensate and a 9–11 $\times$ larger effect on coherence.

What was reported, not predicted (I5). No base relaxes within $t = 1500$: late-window condensates V/P are 0.29/0.89, 0.085/0.64, 0.061/0.66. At $e = 0.60$ the eight imprinted vortices survive the entire run with zero annihilations, $\langle N_v \rangle = 8.0$ in every block: at $T = 0.1$, pairs of separation $L/4$ are protected for at least 1500 time units, Theorem 2 seen directly. In the last three blocks the condensate recovers ($0.29 \rightarrow 0.45$) while N_v stays 8 and Q falls ($0.88 \rightarrow 0.67$): the dipoles contract, the large-scale winding changes and the count does not – thought experiment B in the data. The block-by-block correlation of Δc with ΔN_v is -0.26 and $+0.58$ on the two $e = 0.90$ bases: the condensate follows the arrangement, not the number. The transverse-current stiffness estimator is invalid in V ($n_n > n$), as it was in round 1 whenever free vortices were present.

5.3 The jump and finite size

Round 1 of this programme located the stiffness jump between two quench-started energies one grid step apart, and its equilibrium filter had let through states that a later check showed were

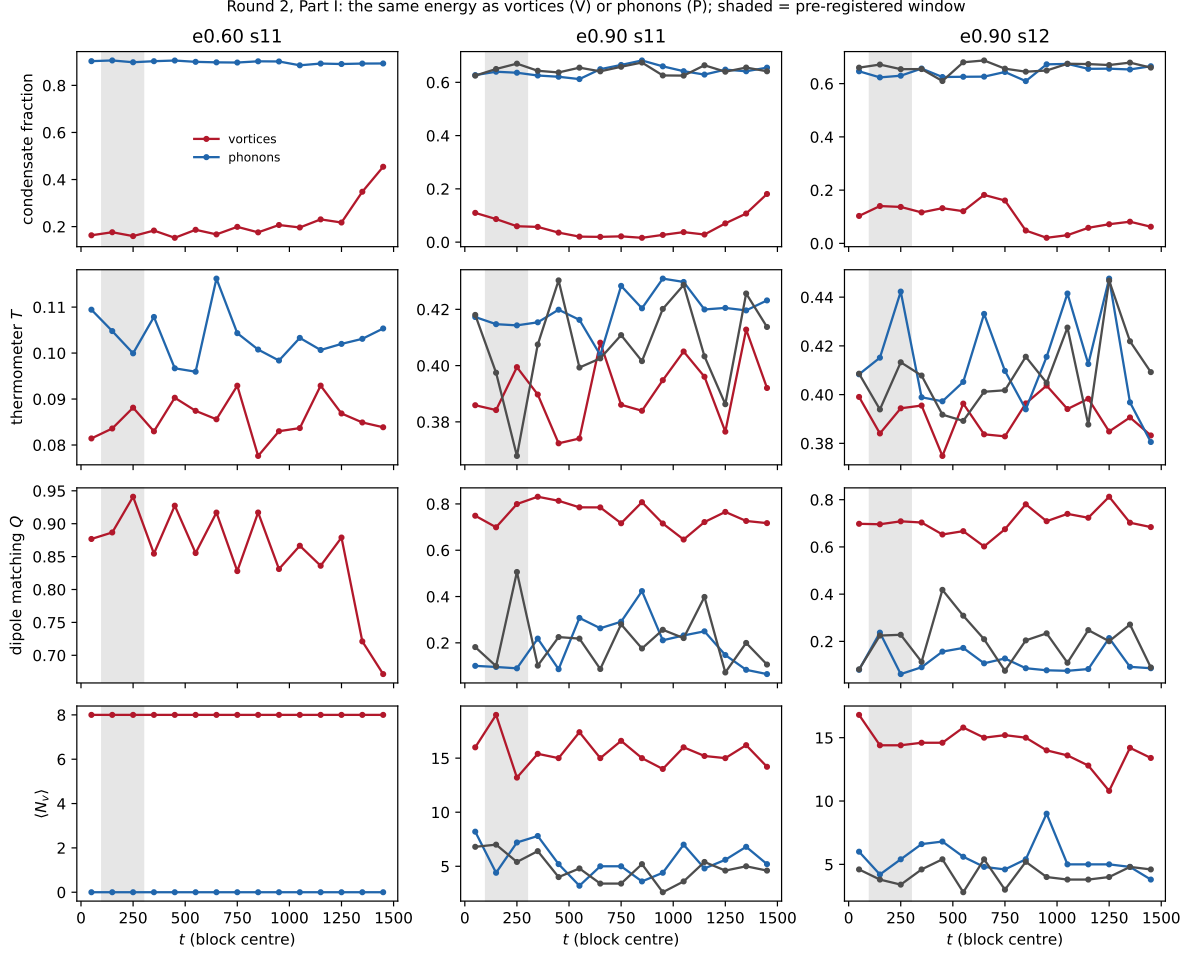


Figure 1: Block observables of the three arms on the three bases. The condensate collapses in V and is untouched in P; the thermometer reads V *below* P; the dipole statistic separates the arms; at $e = 0.60$ the eight imprinted vortices survive the entire run without one annihilation.

not relaxed. Round 2 therefore approaches the jump from the ordered side: the two $e = 0.90$ bases are heated by the phonon construction to $e = 1.00, \dots, 1.20$ in steps of 0.05 (pre-registered) and, after the run log showed the crossing still above $e = 1.20$, to $1.25, \dots, 1.40$ (amendment, made before any verdict was computed). All eighteen runs pass the equilibrium admission. A second box, $L = 32$, is run from quench starts to $t = 4000$ at five energies, three seeds each, all admitted.

The C5 failure is itself a result of the kind this paper is about. The quench state and the ladder state at $e = 1.20$ carry the same number of vortices, about ninety, and the quench state has half the stiffness: the arrangement, not the count, sets n_s . Round 1's transition temperature, 0.72, was read off a quench state and is withdrawn in favour of 0.821. With that correction the Prokof'ev–Ruebenacker–Svistunov number $n\lambda_T^2$ at the transition is 7.65 against $\ln 380 = 5.94$, a 29% excess (round 1 had 47%); the cutoff dependence that would explain it is still not measured. The duality product sits 12–27% above one on all eleven admitted energies of the two boxes, a systematic offset we report and do not explain.

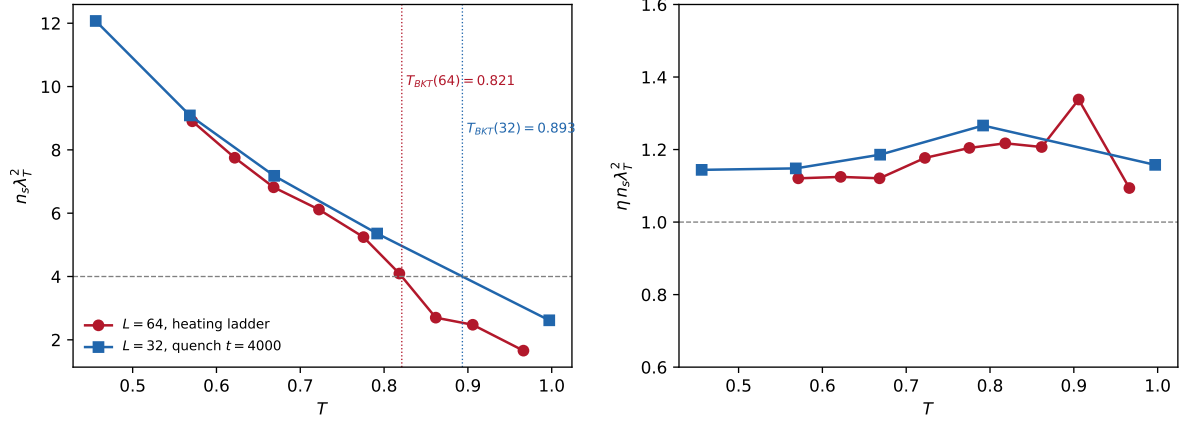


Figure 2: Left: $n_s \lambda_T^2$ against the measured T on the $L = 64$ heating ladder and in the $L = 32$ box, with the Nelson–Kosterlitz level 4 and the two interpolated crossings. Right: the spin-wave duality product $\eta n_s \lambda_T^2$, predicted 1, on the same runs.

statement	result
C5 ladder $e = 1.20$ agrees with round 1's quench $e = 1.20$	fail: $n_s \lambda^2$ 5.24 vs 3.04 at equal $N_v \approx 90$
L1 $n_s \lambda^2 = 4$ crossing inside the ladder	pass: $T_{\text{BKT}}(L=64) = 0.821$, between $e = 1.25$ and 1.30
L2 $\eta = 0.25 \pm 0.08$ at the crossing	pass: 0.307
L3 $\eta n_s \lambda^2 \in [0.75, 1.33]$ wherever $n_s \lambda^2 > 4$	pass: 1.12–1.22 on six energies
F1 $T_{\text{BKT}}(32) \geq T_{\text{BKT}}(64)$	pass: 0.893 $\geq$ 0.821
F2 the drop is sharper in the larger box	pass: $ d(n_s \lambda^2)/dT = 32$ vs 13

Table 2: Pre-registered verdicts of the ladder and the finite-size box.

6 What this establishes, and what it does not

Established.

- The winding number of a sampled phase loop is an integer, is conserved by every continuous evolution without a phase slip, costs a positive barrier to change on an XY ring, is additive over scale so that a bound pair is invisible from outside, and equals the continuum degree for every fine enough sampling. These are theorems.
- At fixed E , N and P , on three equilibrated bases, four injected vortex pairs lower the condensate by 0.49–0.73 and raise η by a factor 9–44; the same energy as phonons does neither. The vortex arm is colder, not warmer, so the difference is not a temperature effect.
- Eight imprinted vortices at $T = 0.1$ survive 1500 time units without one annihilation.
- In the ordered phase, $\eta n_s \lambda_T^2 = 1$ holds to 12–27% on eleven energies in two boxes; the jump sits at $T_{\text{BKT}} = 0.821$ ($L = 64$) with $\eta = 0.31$ there, and moves up and broadens in the smaller box.

Not established.

- The pre-registered composite claim, because its thermometer clause failed on every base. The reading that replaces it – the topological configuration cools the bath, and the difference-making survives against the temperature gradient – is post hoc.
- Why the bath cools. Two mechanisms were tested; one is refuted, one inconclusive; the third is an energy-conservation inference, not a measurement.
- Any transfer of the coupling (link 4) outside this system; the argument principle; a quantitative sampling threshold; the cutoff dependence of the Prokof'ev number; the 12–27% offset of the duality product.
- Statistics. Three bases for the intervention, two seeds per ladder energy. Verdicts are per base; nothing here is a p -value.

Process record. Four amendments: a control redesigned after it proved insensitive (the vortex count cannot see a periodicity defect; a boundary phase jump can, and gave 0.294524 against the predicted 0.294524); a window regranulated before any output; the I2 reading fixed before the data; the ladder widened after the log and before the verdicts. One pre-registered criterion failed in a way none of the fixed readings anticipated, and one criterion showed that the previous round's transition temperature was biased. Both are reported as what they are.

7 What transfers beyond the Bose gas, and what does not

The mathematics of Section 4 never mentions a superfluid. Its objects are a $U(1)$ -valued phase on a loop, a lattice of plaquettes, and a continuous closed loop of angles; its only structural input is $\pi_1(U(1)) = \mathbb{Z}$. The chain integrality $\rightarrow$ conservation by continuity $\rightarrow$ barrier $\rightarrow$ scale additivity $\rightarrow$ continuum limit therefore holds, at the level of description where it is proved, for any $U(1)$ phase field: a superfluid order parameter, but equally a cosmological phase – an axion or Peccei–Quinn field, the phase of a $U(1)$ cosmic-string condensate. Conservation comes from the connectivity of the target space, not from the Hamiltonian, and that is exactly what makes the statement system-independent.

The method transfers too, and it is the more important export: a causal claim about a topological label needs an equally arbitrary control arm – the same energy surplus, delivered in a different form. Without a matched pair, “topology influences” collapses back to the descriptor sense.

What does not transfer is link 4. The coupling of the integer to the observables is empirical, it is measured here in one system with three bases, and a V/P result in a Bose gas says nothing about the coupling in a cosmological model. An astrophysical claim would need its own matched intervention, inside its own model, under its own pre-registered criteria. The honest use of quantum fluids in that programme is as the laboratory of the method – the matched pair, the effect per unit energy, the proved chain – not as a source of numbers.

8 Literature status

Nothing in the physics is new. Phase slips are Anderson's [7]; the ring barrier is Langer and Fisher's [8]; the scale-resolved picture is Kosterlitz and Thouless's [4]; the universal jump is Nelson and Kosterlitz's [9]; Laughlin's argument [6] and Tonomura's experiment [5] are textbook; persistent homology has been used to locate the XY transition [13]; the classical-field method is

Simula and Blakie’s [11]. What this report adds is the chain: each link proved by a kernel or measured under a pre-registered criterion, and an explicit list of gaps.

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